

Kathmandu Engineering College
Department of Electronics, Communication and Information



Sampling and Reconstruction of Signal

Lab Report 2

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1 Objective

- Observe, simulate, and analyze the mathematical process of continuous-time signal discretization, evaluating the criteria for perfect signal reconstruction under the Nyquist-Shannon sampling theorem.

2 Theory

2.1 Introduction to Sampling Theorem

The process of transforming an analog signal into a digital format is designated as **sampling**, which is fundamentally performed by an **analog-to-digital converter (ADC)**. Following discretizations, the signal undergoes secondary transformations via **quantization** and **encoding**.

In the sampling process, a **continuous-time signal** is transformed into an equivalent **discrete-time signal** by evaluating its instantaneous values at periodic intervals. The selection of this periodic timing interval (the sampling rate) must remain sufficiently small to ensure that the analog wave can be recovered from its constituent samples with perfect accuracy. The Sampling Theorem establishes the formal mathematical boundaries for calculating this minimal rate limit.

2.1.1 Sampling Theorem Definition

In general engineering terms, the sampling theorem dictates that:

“An analog signal can be perfectly reproduced from an appropriate set of its periodically spaced samples if the sampling frequency exceeds twice the highest frequency component contained within the signal.”

This theorem underpins all digital communication systems, making it viable to transmit discrete code-word indices over band-limited channels instead of resource-heavy continuous waveforms.

The theorem is broken down into two core operating criteria:

- **Transmission Limit:** A finite-energy, band-limited signal containing no frequency components higher than f_m Hz is completely specified by discrete values spaced uniformly at periods of $T_s = \frac{1}{2f_m}$ seconds.
- **Reconstruction Limit:** A finite-energy, band-limited signal capped at f_m Hz can be perfectly reconstructed from its samples if it is evaluated at a minimum Nyquist sampling rate of $f_s \geq 2f_m$ samples per second.

2.1.2 Mathematical Proof of Sampling

Consider an analog signal $x(t)$ whose spectral bandwidth is limited to f_m Hz. Symmetrical uniform sampling is realized by modulating $x(t)$ with a periodic impulse train $\delta_{T_s}(t)$, yielding the idealized sampled output sequence $x_\delta(t)$.

The resulting discrete signal features individual impulses spaced at intervals of T_s seconds, with scaling amplitudes corresponding directly to $x(nT_s)$, where $n = 0, \pm 1, \pm 2, \dots$. The term T_s denotes the sampling period, and its mathematical reciprocal, $f_s = \frac{1}{T_s}$, defines the sampling frequency. The transformation is modeled as:

$$x_\delta(t) = x(t) \cdot \delta_{T_s}(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \delta(t - nT_s) \quad (1)$$

Where $\delta(t - nT_s)$ represents the **Dirac delta** function offset precisely to the discrete time coordinates $t = nT_s$. In equation (1), each delta function in the series is weighted by the corresponding sample value of the input signal $x(t)$, i.e., $x(nT_s)$.

As the impulse train $\delta_{T_s}(t)$ is a periodic signal with period T_s , it can be expressed as an exponential Fourier series, as shown by Eq. (2):

$$\delta_{T_s}(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn2\pi f_s t} \quad (2)$$

Where $C_n = 1/T_s$,

$$\delta_{T_s}(t) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} e^{jn2\pi f_s t} \quad (3)$$

Therefore,

$$x_\delta(t) = x(t) \delta_{T_s}(t) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} x(t) e^{jn2\pi f_s t} \quad (4)$$

To find $X_\delta(f)$, the Fourier transform of $x_\delta(t)$, we take the Fourier transform of the summation in the previous relation. Based on the frequency-shifting property, the transform of the n^{th} term is shifted by nf_s .

$$\begin{aligned} x(t) &\xleftrightarrow{\text{FT}} X(f) \\ x(t) e^{jn2\pi f_s t} &\xleftrightarrow{\text{FT}} X(f - nf_s) \end{aligned}$$

Therefore,

$$X_\delta(f) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} X(f - nf_s) \quad (5)$$

This means that the spectrum $X_\delta(f)$ consists of $X(f)$ (Spectrum of signal $x(t)$) scaled by a constant $1/T_s$, repeating periodically with period $f_s = 1/T_s$ Hz, as shown in Fig. 1(f).

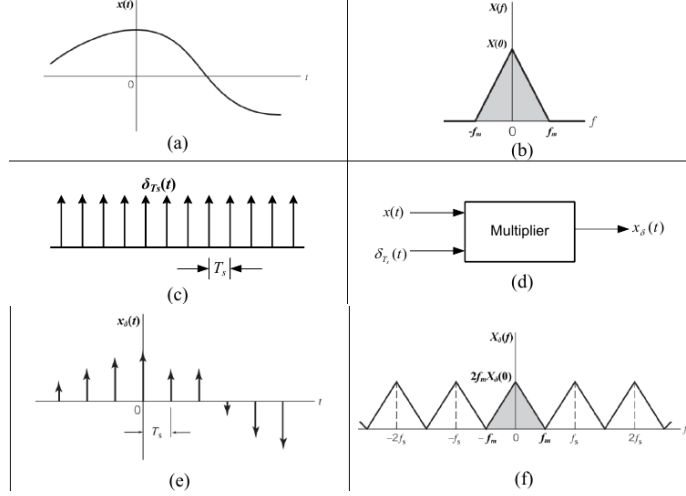


Figure 1: Signal discretization steps: (a) Continuous signal $x(t)$, (b) Baseband spectrum of $x(t)$, (c) Impulse train modulation function, (d) Multiplier model block, (e) Resulting sampled signal $x_\delta(t)$, and (f) Periodic spectrum of the sampled signal.

If we reconstruct $x(t)$ from $x_\delta(t)$, we should be able to recover $X(f)$ from $X_\delta(f)$. This is possible if there is no overlap between successive cycles of $X_\delta(f)$. Fig. 1(f) shows that this requires

$$f_s \geq 2f_m \quad (6)$$

Also, the sampling interval $T_s = 1/f_s$. Therefore,

$$T_s \leq \frac{1}{2f_m} \quad (7)$$

Thus, as long as the sampling frequency f_s **is greater than twice the signal bandwidth f_m (in hertz, Hz)**, $X_\delta(f)$ will consist of non-overlapping repetitions of $X(f)$. When this is true, Fig. 1(f) shows that $x(t)$ can be recovered from its samples by passing the sampled signal $x_\delta(t)$ through an ideal low-pass filter of bandwidth f_m Hz. The minimum sampling rate $f_s = 2f_m$ required to recover $x(t)$ from its samples $x_\delta(t)$ is called the **Nyquist rate for $x(t)$** , and the corresponding sampling interval $T_s = 1/2f_m$ is called the **Nyquist interval for $x(t)$** , given by Eq. (6) and Eq. (7) respectively.

2.1.3 Signal Reconstruction

The process of reconstructing a continuous time signal $x(t)$ from its samples is known as **interpolation**. Fig. 1 shows the constructive proof that a signal $x(t)$ band-limited to f_m Hz, can be reconstructed (interpolated) exactly from its samples. This is done by passing through an ideal low-pass filter of bandwidth f_m Hz.

The expression for the sampled signal is written as:

$$x_\delta(t) = x(t)\delta_{T_s}(t) \quad (8)$$

As seen from Eq. (1), the sampled signal contains a component $(1/T_s)x(t)$ when $n = 0$, and to recover $x(t)$ [or $X(f)$], the sampled signal must be sent through an ideal low-pass filter of bandwidth f_m Hz and gain T_s . Such an ideal filter response has the transfer function:

$$H(f) = T_s \times \text{rect} \left(\frac{f}{2f_m} \right) \quad (9)$$

2.1.4 Ideal Reconstruction

To recover the analog signal from its uniform samples, the ideal interpolation filter transfer function found in Eq. (9) is shown in Fig. 2(a).

The impulse response of this filter, i.e., the inverse Fourier Transform of $H(f)$, is:

$$h(t) = \mathcal{F}^{-1} \left[T_s \times \text{rect} \left(\frac{f}{2f_m} \right) \right] = 2f_m T_s \text{sinc}(2f_m t) \quad (10)$$

Assuming that sampling is done at the Nyquist rate, then $T_s = \frac{1}{2f_m}$. So that, $2f_m T_s = 1$.

Substituting the value of $2f_m T_s$ in the previous equation, we have:

$$h(t) = \text{sinc}(2f_m t) \quad (11)$$

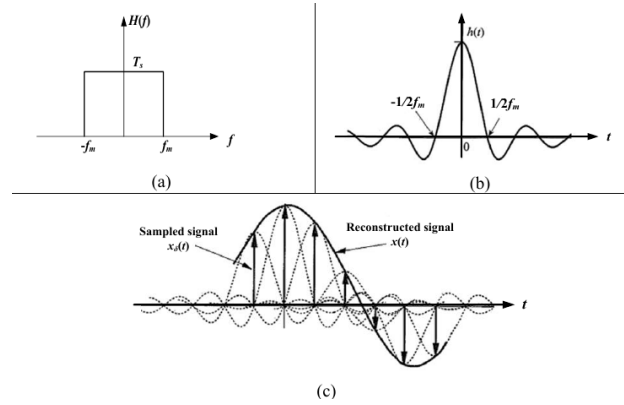


Figure 2: (a) Ideal reconstruction filter. (b) Impulse response of ideal reconstruction. (c) Reconstructed signal.

The impulse response $h(t)$ given in Eq. 11 is shown in fig. 2(b). It can be seen from fig. 2(b) that $h(t)=0$ at all Nyquist sampling instants i.e., at $t = \pm n/(2f_m)$, except at $t=0$.

Each sample in $x_s(t)$, being an impulse, produced a sinc pulse of height equal to the strength of the sample. Addition of the sinc pulses produced by all the sample results in $x(t)$. For instant, the k^{th} sample of the input $x_s(t)$ is the impulse will be $x(kT_s)\delta(t - kT_s)$. Therefore, the filter output to $x_s(t)$ i.e., $x(t)$, may be expressed as a sum given in

$$x(t) = \sum_k x(kT_s)h(t - kT_s) \quad (12)$$

$$x(t) = \sum_k x(kT_s) \text{sinc}[2\pi f_m(t - kT_s)] \quad (13)$$

$$x(t) = \sum_k x(kT_s) \text{sinc}(2\pi f_m t - k\pi) \quad (14)$$

Eq.14 is the interpolation formula, which yields values of $x(t)$ between samples as a weight sum of all the sample values as shown in fig.2(c)

3 Source Code & Simulation Output

3.1 The Original/Sample Signal

```
t1 = 0:0.1:20
f1 = 0.2
x1 = sin(2*pi*f1*t1)

f2 = 0.3
x2 = sin(2*pi*f2*t1)
% Continuous Signal
x = x1 + x2;
figure(1)
subplot(2,1,1)
plot(t1,x)
hold on
plot([0 20],[0 0], 'black')
hold off
title('Original Continuous Signal')
xlabel('time')
ylabel('x[t]')
grid on

% Continuous signal -> Discrete Signal
x_sample = x(1:10:201)
subplot(2,1,2)
plot([0 20],[0 0], 'r')
hold on
stem(x_sample, 'filled')
hold off
title('sample Discrete Signal')
xlabel('time')
ylabel('x_s[n]')
axis([0 20 -2 2])
hold on
grid
```

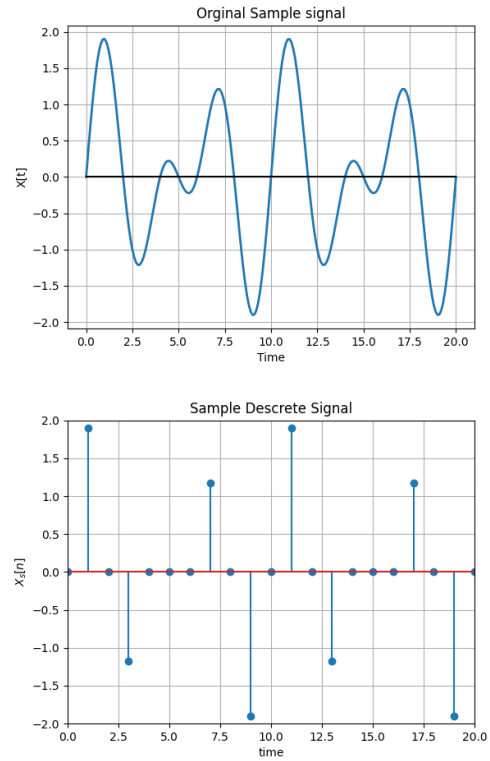


Figure 3: Original Continuous and Discrete Signal.

3.2 Reconstruction Signal

```
figure(2)
subplot(2,1,2)
plot(t1,x,'blue')
hold on
plot([0 20],[0 0])
hold off
title('Continuous Signal')
xlabel('time')
ylabel('x[t]')
grid on

x_recon=0;
subplot(2,1,1);
d = length(x_sample);
for k=0:d-1
    stem(0:d-1,x_sample,'filled');
    if k==d-1
        title('reconstruction finished')
    );
    else
        title('sample by reconstruction
');
    end
    grid on
    l=k:-0.1:-20+k;
    x_recon = x_recon + x_sample(k+1) *
    sinc(l);
    axis([0 20 -2 2]);
    hold on

    plot(t1,x_sample(k+1)*sinc(l));
    plot(t1,x_recon);
    hold off

end
```

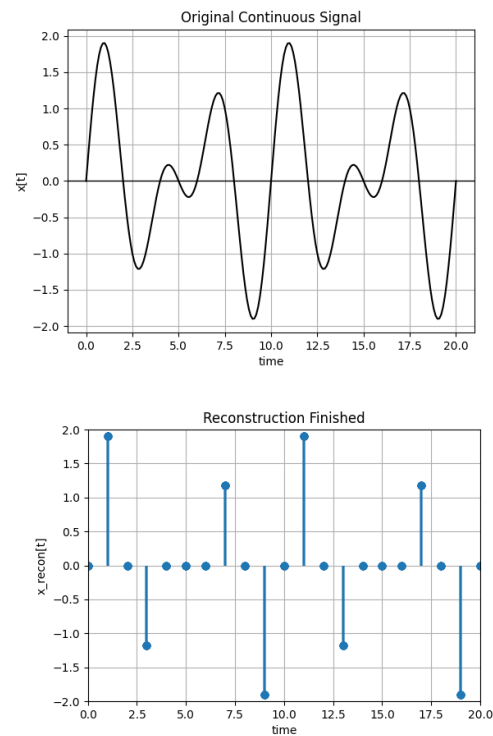


Figure 4: Reconstructed Continuous and Discrete Signal.

4 Discussion

This laboratory assignment concentrated on analyzing the sampling process and verifying the Nyquist criteria using MATLAB simulation scripts. The computational processing highlights several key engineering observations:

- **Spectral Replication:** Modulating an analog signal with a periodic impulse train creates infinite spectral replicas in the frequency domain, spaced at integer multiples of the sampling frequency (f_s).
- **Aliasing Vulnerability:** If the sampling rate falls below the Nyquist threshold ($f_s < 2f_m$), high-frequency spectral replicas overlap with the baseband signal. This overlap introduces aliasing distortion, permanently corrupting the signal's original identity.
- **Ideal Reconstruction Context:** The simulation of the $\text{sinc}(t)$ function highlights its role as an ideal low-pass filtering interpolation kernel. Convolution of a discrete sample sequence with a sinc function accurately recovers the continuous baseband signal by filtering out all higher-order spectral duplicates.

5 Conclusion

In this laboratory experiment, the Nyquist-Shannon Sampling Theorem was successfully studied and verified using MATLAB. The mathematical conversion of continuous signals ($X(t)$) into discrete representations ($X[n]$) was demonstrated alongside the spectral effects of varying the sampling interval (T_s).

The simulation results confirmed that choosing an engineering sample rate of $f_s \geq 2f_m$ is critical to preventing aliasing. Additionally, evaluating the sinc interpolation function confirmed that ideal low-pass filtering can perfectly extract baseband information from an impulse-sampled wave. Ultimately, this experiment provided essential practical experience in signal digitization, establishing the core foundations needed for digital signal processing (DSP) and modern communication networks.

The figures were made using **Python** library for better visualization of the figure. The actual **MatLab** figure are shown below:

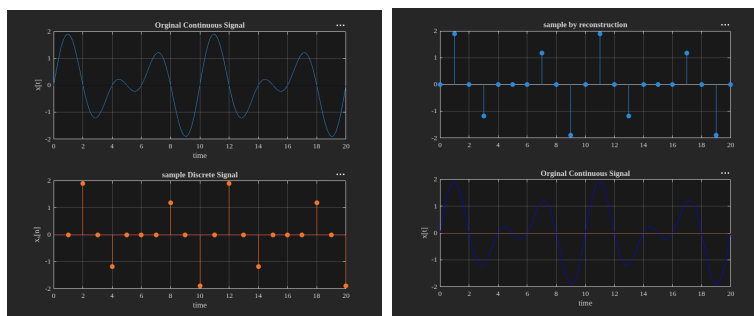


Figure 5: MatLab Original Figures